

Lab07

Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)

con <- dbConnect(SQLite(), "database.sqlite3")
```

Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT DISTINCT i.name AS professor
FROM instructors AS i
JOIN teachings AS t
ON i.id = t.instructor_id
JOIN sections AS s
ON s.uuid = t.section_uuid
JOIN subject_memberships AS sm
ON sm.course_offering_uuid = s.course_offering_uuid
JOIN subjects AS sub
ON sub.code = sm.subject_code
WHERE sub.abbreviation = 'STAT'
ORDER BY i.name;
```

Displaying records 1 - 10

professor

ABIGAIL BENZINE

ADIN-CRISTIAN ANDREI

AKICHIKA OZEKI

ALAN HUANG

ALBERTO DEL PIA

ALBRECHT KLEMM

ALEXANDER COVINGTON

ALEXANDER FISH

ALEXANDER KISELEV

ALYSSA DIGILIO

O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

```
SELECT gd.course_offering_uuid, gd.section_number,
(4.0 * CAST(gd.a_count AS REAL) + 3.5 * CAST(gd.ab_count AS REAL) +
3.0 * CAST(gd.b_count AS REAL) + 2.5 * CAST(gd.bc_count AS REAL) + 2.0 * CAST(gd.c_count AS RE
```

```
(CAST(gd.a_count AS REAL) + CAST(gd.ab_count AS REAL) + CAST(gd.b_count AS REAL) + CAST(gd.bc_count AS REAL)
FROM grade_distributions AS gd
WHERE CAST(gd.a_count AS REAL) + CAST(gd.ab_count AS REAL) + CAST(gd.b_count AS REAL) + CAST(gd.bc_count AS REAL) > 3.5;
```

Displaying records 1 - 10

course_offering_uuid	section_number	GPA
344b3ebe-da7e-314c-83ed-9425269695fd	1	4.000000
f718e6cd-33f0-3c14-a9a6-834d9c3610a8	1	4.000000
ea3b717c-d66b-30dc-8b37-964d9688295f	1	3.910256
075da420-5f49-3dd0-93df-13e3c152e1b1	1	4.000000
2b4e216d-a728-3713-8c7c-19affc6b2fd	1	4.000000
87966a7b-f676-33d0-83d2-acdb67da6790	1	4.000000
ff511882-5eab-3e7b-a89f-8fbfd1906127	1	3.978814
b3e26604-fe6e-30df-bc91-b5fa57717a7e	1	3.957944
e100d196-5e82-32e4-80e9-ac45c07a498c	1	3.977477
76b9c458-d3c2-38c4-951f-69b6900bd7fe	1	3.921986

Quem é o professor mais difícil?

```
SELECT i.name AS professor,
AVG((4.0 * CAST(gd.a_count AS REAL) + 3.5 * CAST(gd.ab_count AS REAL) + 3.0 * CAST(gd.b_count AS REAL) + CAST(gd.bc_count AS REAL)
FROM instructors AS i
JOIN teachings AS t
  ON i.id = t.instructor_id
JOIN sections AS s
  ON s.uuid = t.section_uuid
JOIN subject_memberships AS sm
  ON sm.course_offering_uuid = s.course_offering_uuid
JOIN subjects AS sub
  ON sub.code = sm.subject_code
JOIN grade_distributions AS gd
  ON gd.course_offering_uuid = s.course_offering_uuid
  AND gd.section_number = s.number

WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
ORDER BY GPA
LIMIT 1;
```

1 records

professor	GPA
CHRISTINE SORKNESS	NA

Quem é o professor mais fácil?

```
SELECT i.name AS professor,
AVG((4.0 * CAST(gd.a_count AS REAL) + 3.5 * CAST(gd.ab_count AS REAL) + 3.0 * CAST(gd.b_count AS REAL) + 2.0 * CAST(gd.bc_count AS REAL) + 1.0 * CAST(gd.cd_count AS REAL)) /
(CAST(gd.a_count AS REAL) + CAST(gd.ab_count AS REAL) + CAST(gd.b_count AS REAL) + CAST(gd.bc_count AS REAL) + CAST(gd.cd_count AS REAL))
FROM instructors AS i
JOIN teachings AS t
    ON i.id = t.instructor_id
JOIN sections AS s
    ON s.uuid = t.section_uuid
JOIN subject_memberships AS sm
    ON sm.course_offering_uuid = s.course_offering_uuid
JOIN subjects AS sub
    ON sub.code = sm.subject_code
JOIN grade_distributions AS gd
    ON gd.course_offering_uuid = s.course_offering_uuid
    AND gd.section_number = s.number

WHERE sub.abbreviation = 'STAT'
GROUP BY i.id, i.name
ORDER BY GPA DESC
LIMIT 1;
```

1 records

professor	GPA
SRIKANTHMADHAVAN ARAVAMUTHAN	4

Qual é a disciplina mais difícil e mais fácil

```
SELECT sub.abbreviation AS assunto, c.number AS disciplina, c.name AS nome,
AVG((4.0 * CAST(gd.a_count AS REAL) + 3.5 * CAST(gd.ab_count AS REAL) + 3.0 * CAST(gd.b_count AS REAL) + 2.0 * CAST(gd.bc_count AS REAL) + 1.0 * CAST(gd.cd_count AS REAL)) /
(CAST(gd.a_count AS REAL) + CAST(gd.ab_count AS REAL) + CAST(gd.b_count AS REAL) + CAST(gd.bc_count AS REAL) + CAST(gd.cd_count AS REAL))
FROM subjects AS sub
JOIN subject_memberships AS sm
    ON sub.code = sm.subject_code
JOIN course_offerings AS co
    ON co.uuid = sm.course_offering_uuid
JOIN courses AS c
    ON c.uuid = co.course_uuid
JOIN grade_distributions AS gd
    ON gd.course_offering_uuid = co.uuid
WHERE sub.abbreviation = 'STAT'
GROUP BY c.uuid, c.number, c.name
ORDER BY GPA;
```

Displaying records 1 - 10

assunto	disciplina	nome	GPA
STAT	698	Directed Study	NA
STAT	546	Practicum in Clinical Trial Data Analysis and Interpretation	NA
STAT	990	Research	NA
STAT	768	Statistical Methods for Medical Image Analysis	NA
STAT	699	Directed Study	NA
STAT	681	Senior Honors Thesis	NA
STAT	682	Senior Honors Thesis	NA
STAT	431	Introduction to the Theory of Probability	2.891568
STAT	525	Linear Programming Methods	2.957360
STAT	309	Introduction to Probability and Mathematical Statistics I	3.031951

```
dbDisconnect(con)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-22 17:06:33 -03"
```